

PhotoGraphiQ

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PhotoGraphiQ is an open-source Python framework for continuous-variable photonic measurement-based quantum computing. It separates weighted graph resources, measurement patterns, real-valued classical dependencies, Gaussian compilation, and physical simulation. **Piquasso** supplies photonic gate execution; an independent phase-space backend and an affine channel analyzer provide complementary numerical checks. The framework supports finite squeezing, adaptive quadrature measurements, general Gaussian measurements, displacement feed-forward, causal validation, numerical CV-flow certificates for specified measurement orders, serialization, and visualization. A Gaussian compiler constructs teleportation patterns for single-mode symplectic gates and two-mode interactions, while an experimental Fock adapter supports correlated resources, conditional photon counting and homodyne, Kerr and cubic gates, ideal ladder operations, and finite cat/cubic resource injection with nonlinear feed-forward and cutoff diagnostics. We derive the conventions, measurement updates, teleportation noise, compiler factorization, and displacement-frame rules used by the software. Validation distinguishes analytical physics, raw **Piquasso** comparisons, and structural comparisons with the qubit MBQC framework **GraphiX**. Reproducible finite-squeezing data illustrate the difference between conditional trajectories and unconditional channels. The release targets transparent, testable Gaussian MBQC and resource experiments; it does not claim universal non-Gaussian compilation, automatic differentiation, or fault-tolerant photonic computation.

1 Introduction

Measurement-based quantum computing (MBQC) encodes a computation in an entangled resource and a sequence of measurements and classical corrections. In the continuous-variable (CV) setting, bosonic modes replace qubits, quadrature measurements produce real-valued outcomes, and physical graph resources have finite squeezing. These differences affect both the mathematics of computation and the structure of a useful software interface. CV cluster computation and its Gaussian and non-Gaussian resources have well-established theoretical foundations [4, 3]. A software implementation must make its finite-resource and convention choices explicit rather than inherit binary measurement rules from a qubit implementation.

Piquasso provides photonic circuit simulation in Gaussian and Fock representations [1]. **GraphiX** provides an architectural reference for qubit measurement patterns, command manipulation and simulation [2]. **PhotoGraphiQ** occupies the layer between these capabilities: its primary objects are graph resources and adaptive measurement computations, and its backend translates their physical operations into photonic simulation. The intended extension path is **Piquasso** $\rightarrow$ **PhotoGraphiQ** $\rightarrow$ a future **PhotoGraphiQML** package. No machine-learning dependencies are introduced in the present runtime.

1.1 Contributions and assumptions

This article describes the implemented version 0.2.0 repository. Its contributions are a backend-independent command representation with causal classical signals; finite-squeezing Gaussian graph preparation and exact destructive measurement conditioning; a teleportation-based Gaussian compiler; independent unconditional channel analysis; specified-order CV-flow certification; a conservative displacement

frame and rewrite system; and a validation suite with reproducible examples and data. The non-Gaussian extension adds finite-resource injection, conditional wavefunction updates, physical heralding, occupation-aligned state metrics and independent Fock-space validation.

The assumptions are explicit. Gaussian trajectories use dense first and second moments with $\hbar = 2$. Gaussian compilation implements noisy finite-resource channels whose ideal symplectic targets are checked separately. The Fock adapter uses a global total-photon cutoff and supports ideal adaptive homodyne conditioning through an explicit wavefunction adapter. CV-flow is checked for a supplied total order; the software does not search for maximally parallel flow. A dependency DAG by itself is not a determinism certificate. GraphiX validates corresponding structures, not equality between CV and qubit states.

The work is a software implementation and validation report. It does not claim a new teleportation identity, new Gaussian compilation theorem, experimental photonic data, or a performance advantage over existing simulation libraries. The manuscript is a draft for author review, not a statement of publication or acceptance by *Quantum*.

2 Architecture and computational objects

The package uses a `src/photographiq` layout. Graph topology, commands, expressions and measurement descriptions do not contain Piquasso instructions. The execution engine resolves classical values and calls an abstract backend. Backend state is kept separate from the pattern, so the same computation can be executed repeatedly with different inputs, parameters or random seeds.

2.1 Graphs and node roles

`CVGraph` wraps an undirected simple weighted graph. Vertices have stable labels and resource squeezing, while edges have real coupling weights. Input and output tuples preserve user ordering. Ancillas are non-input nodes; measured nodes are identified by a pattern’s measurement commands. Directed resources, multiedges, self-loops and nonfinite numerical weights are rejected. A

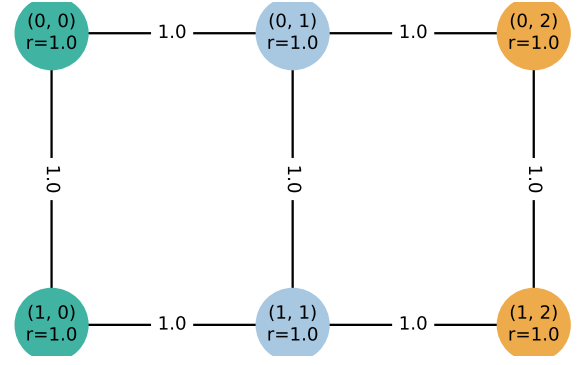


Figure 1: An exported two-row resource. Labels, squeezing and weights belong to the resource description. Green and amber nodes denote inputs and outputs; measurements and dependencies are added at the pattern layer.

diagonal quadratic phase is a local operation rather than a self-edge in this resource model.

Constructors cover lines, rings, stars, trees, rectangular and square lattices, adjacency matrices and arbitrary NetworkX graphs. Temporal resources currently mean time-labelled line topology, and the dual-rail constructor means a ladder graph. These names do not assert that an experimental time-multiplexed interferometer has been synthesized. Figure 1 illustrates the labelled-resource view.

2.2 Patterns, commands and execution

The command vocabulary consists of preparation, entanglement, measurement, displacement, rotation, squeezing, beam splitter, loss, cubic phase, classical signal and output declaration. A pattern can be constructed from a graph or by appending commands. Input modes are supplied separately and cannot be prepared again. A measurement removes its mode and writes a result under a unique key; the default key is the node label. An optional final output declaration can select or reorder survivors, tracing out other modes.

The execution sequence is

$$\begin{aligned} \text{command} &\rightarrow \text{resolve values} \rightarrow \text{backend} \\ &\rightarrow \text{record outcome} \rightarrow \text{next command.} \end{aligned} \quad (1)$$

This is a genuinely adaptive sequence: later measurements may depend on earlier real outcomes. **Result** stores measurement outcomes, all classical records, an independent output-state snapshot, backend identity and seed information. The pattern itself is not mutated by simulation.

Layer	Principal modules	Responsibility
Resource	graph, states	Weighted topology, node roles, Gaussian/Fock inputs
Intermediate representation	commands, pattern	Ordered physical and classical instructions
Measurement calculus	measurements, expressions	POVM descriptions, parameter and outcome dependencies
Causality and flow	flow, cvflow	Command scheduling and distinct real-linear flow certificates
Compilation	compiler, protocols	Teleportation gadgets and Gaussian circuit translation
Execution	simulator, backends	Conditional trajectories, frames and native gates
Analysis	gaussian, analysis	Symplectic identities and unconditional Gaussian channels
Interchange	serialization, visualization	Allowlisted JSON and resource/dependency figures
Research artifacts	tests, examples, experiments, docs, paper	Validation, tutorials, data and manuscript

Table 1: Repository organization. The Gaussian measurement adapter is shared by the two Gaussian execution backends; the independent channel analyzer and analytical tests supply additional validation.

2.3 Expression safety and causality

An external **Parameter** and a previous **Outcome** are distinct expression leaves. Arithmetic and a small set of elementary functions construct an expression tree. Runtime resolution requires finite real scalar values. Vector-valued general-dyne outcomes are accessed by an explicit component. **Signal** commands create named intermediate classical registers.

Callables are permitted through a wrapper with declared dependencies. The function receives only those records, so undeclared access fails even if the record happens to exist. Callables cannot be serialized. The portable expression language uses neither Python string evaluation nor arbitrary code deserialization.

Causal validation creates one graph vertex per command, inserts producer-to-user edges for classical records, and preserves consecutive touches of each quantum mode. It rejects missing producers, duplicate keys, cycles, future dependencies, double preparation, and operations on measured modes. Topological scheduling then provides a legal command order under these constraints. It is not a search for a different measurement calculus or a proof of ideal-limit convergence.

3 Phase-space conventions

The internal quadrature vector and commutators are

$$\mathbf{R} = (q_0, p_0, q_1, p_1, \dots)^T, \quad [R_j, R_k] = 2i\Omega_{jk}, \quad (2)$$

where $\Omega = \bigoplus_j \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Means and statistical covariance are

$$\boldsymbol{\mu} = \langle \mathbf{R} \rangle, \quad V_{jk} = \frac{1}{2} \langle \{\Delta R_j, \Delta R_k\} \rangle. \quad (3)$$

Vacuum has $V = \mathbb{I}$ and Gaussian physicality requires $V + i\Omega \succeq 0$. The state validator checks dimensions, labels, finiteness, symmetry and this uncertainty condition with a documented numerical tolerance.

Piquasso’s covariance convention is the unhalved anticommutator matrix $\sigma = 2V$. Thus native covariance and displacement units are converted at the backend boundary. The real symplectic matrices for rotation and squeezing are

$$R(\phi) = \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix}, \quad (4)$$

$$S(r) = \text{diag}(e^{-r}, e^r). \quad (5)$$

Positive gate squeezing reduces q variance. Positive *resource* squeezing r instead means preparation with $S(-r)$, reducing p variance.

The controlled-phase and translation conventions are

$$CZ_{ij}(g) = e^{igq_i q_j/2}, \quad (6)$$

$$X(s) = e^{-isp/2}, \quad Z(t) = e^{itq/2}. \quad (7)$$

Equation (6) sends $p_i \mapsto p_i + gq_j$ and $p_j \mapsto p_j + gq_i$. The translations shift q and p by s and t ; Piquasso's complex displacement is therefore $\alpha = (s + it)/2$. No angle in the public interface is implicitly measured in units of π .

4 Finite-squeezing graph resources

For a symmetric zero-diagonal adjacency matrix A , prepare independent momentum-squeezed modes and apply the product of weighted CZ gates. In grouped coordinate order $(\mathbf{q}, \mathbf{p})$, let

$$D_q = \text{diag}(e^{2r_j}), \quad D_p = \text{diag}(e^{-2r_j}). \quad (8)$$

The graph symplectic map and covariance are

$$S_A = \begin{pmatrix} \mathbb{I} & 0 \\ A & \mathbb{I} \end{pmatrix}, \quad (9)$$

$$V_A = \begin{pmatrix} D_q & D_q A \\ AD_q & D_p + AD_q A \end{pmatrix}. \quad (10)$$

The implementation stores these matrices in interleaved order.

Proposition 1. *The nullifiers $\delta = \mathbf{p} - A\mathbf{q}$ of the resource in Eq. (10) commute and have covariance D_p .*

Proof. Symmetry of A cancels the commutators between the two linear terms. Under S_A , $\mathbf{p} - A\mathbf{q}$ is the original independent squeezed momentum vector. Its covariance is consequently D_p . $\square$

The nullifier identity provides a sensitive test of edge signs, squeezing axes, coordinate order and covariance factors. Arbitrary input states can replace the designated input preparations, including correlated Gaussian inputs. The simple resource-only nullifier covariance must then be interpreted with the actual input state rather than reused without qualification.

An ideal symbolic graph retains topology and nullifier coefficients but cannot be executed as a finite covariance state. Infinite squeezing is not encoded by overflowing floating-point values. Piquasso's native `Graph` preparation is not used as a substitute for the canonical resource in Eq. (10).

5 Measurements and classical adaptation

5.1 Exact homodyne conditioning

Partition the measured mode and the survivors as

$$\boldsymbol{\mu} = \begin{pmatrix} \mathbf{b} \\ \mathbf{a} \end{pmatrix}, \quad V = \begin{pmatrix} B & C^T \\ C & A \end{pmatrix}. \quad (11)$$

For $h = (\cos \theta, \sin \theta)$, homodyne measures $y = h\mathbf{R}_m$. With nonnegative calibrated detector noise ν , define

$$v = hBh^T + \nu, \quad y \sim \mathcal{N}(h\mathbf{b}, v). \quad (12)$$

The conditional survivor state is

$$\boldsymbol{\mu}' = \mathbf{a} + Ch^T \frac{y - h\mathbf{b}}{v}, \quad (13)$$

$$V' = A - \frac{Ch^T h C^T}{v}. \quad (14)$$

An ideal detector uses $\nu = 0$. Efficiency $0 < \eta \leq 1$ is modeled by vacuum mixing followed by calibration to the incoming quadrature, yielding $\nu = (1 - \eta)/\eta + \nu_{\text{readout}}$. The measured mode is removed. Linear solves are used for matrix-valued conditioning; singular or invalid requests raise errors.

5.2 Heterodyne and general Gaussian measurements

A general-dyne detector is represented by a physical statistical covariance M , with $M + i\Omega_1 \succeq 0$. The outcome and update are

$$\mathbf{y} \sim \mathcal{N}(\mathbf{b}, B + M), \quad (15)$$

$$\boldsymbol{\mu}' = \mathbf{a} + C(B + M)^{-1}(\mathbf{y} - \mathbf{b}), \quad (16)$$

$$V' = A - C(B + M)^{-1}C^T. \quad (17)$$

Heterodyne uses $M = \mathbb{I}$. It returns two phase-space coordinates, so its vacuum outcome covariance is $2\mathbb{I}$ in the chosen units. These values should not be confused with the real and imaginary parts of $\alpha = (q + ip)/2$.

5.3 Native-backend convention audit

Source inspection and tests exposed differences that are recorded in `docs/design.md`. The inspected Piquasso ControlledZ Bogoliubov blocks produce the positive momentum update used

here, despite an opposite sign in its displayed unitary docstring. Its Gaussian homodyne routine uses a finite detector covariance $\text{diag}(z^2, z^{-2})$, with q homodyne approached as $z \rightarrow 0$.

More significantly, the pinned version 8.0.1 passes an anticommutator covariance to its Gaussian random sampler without the statistical factor $1/2$. A separate raw characterization test consequently observes vacuum heterodyne variance near 4 rather than 2. PhotoGraphiQ does not silently inherit that sampling convention. Piquasso performs native gates; PhotoGraphiQ performs the exact Gaussian measurement adapter in Eqs. (13)–(14) and their general-dyne counterparts. Native conditional moments can still be checked at native sample values; this tests the state update, not the physical sampling distribution. Dependency versions are pinned so such behavior remains reviewable.

6 Teleportation protocols and Gaussian compilation

6.1 Elementary cluster step

Take an input mode and an independent momentum-squeezed ancilla. Apply unit CZ, measure $m = p_0 + kq_0$ on the input, and correct the ancilla by $X(-m)$. In terms of the pre-entanglement variables, the resulting output relations are

$$\begin{aligned} \mathbf{R}_{\text{out}} &= T(k)\mathbf{R}_{\text{in}} + (0, p_{\text{anc}})^T, \\ T(k) &= \begin{pmatrix} -k & -1 \\ 1 & 0 \end{pmatrix}. \end{aligned} \quad (18)$$

Indeed, before the measurement the measured momentum is $p_0 + q_1$, while the surviving momentum is $p_1 + q_0$. Subtracting the measured value from the surviving position cancels q_1 .

The physical measurement description is normalized: the software sets $\theta = \text{atan2}(1, k)$ and records $y = q \cos \theta + p \sin \theta$. The feed-forward value is $m = y / \sin \theta$, not y . This normalization is necessary for nonzero k .

Proposition 2. *For Gaussian inputs, the outcome-averaged corrected channel of a cluster step is*

$$\begin{aligned} \boldsymbol{\mu} &\mapsto T(k)\boldsymbol{\mu}, \\ V &\mapsto T(k)VT(k)^T + N_r, \\ N_r &= \text{diag}(0, e^{-2r}). \end{aligned} \quad (19)$$

Proof. The input and ancilla variables in Eq. (18) are initially independent. The ancilla momentum has zero mean and variance e^{-2r} . Taking unconditional first and second moments gives Eq. (19). Equivalently, Gaussian Wigner-variable propagation integrates the homodyne-conditioned branches. $\square$

This proposition is deliberately an unconditional statement. Finite-resource conditional means and covariances depend on the observed branch. Averaging only conditional covariance omits the covariance of the conditional means.

6.2 Wires, identity and optical teleportation

A single $k = 0$ step is a Fourier transformation, $T(0) = R(\pi/2)$. Four such steps implement identity with added covariance $2e^{-2r}\mathbb{I}$ for uniform resources. For a general wire, the exact recursion is

$$S_j = T(k_j)S_{j-1}, \quad (20)$$

$$N_j = T(k_j)N_{j-1}T(k_j)^T + N_{r_j}, \quad (21)$$

$$S_0 = \mathbb{I}, \quad N_0 = 0. \quad (22)$$

The repository separately implements Braunstein–Kimble optical teleportation [6]. Oppositely squeezed resources, a balanced beam splitter, complementary Bell homodynes and gain- $\sqrt{2}$ corrections produce $V_{\text{out}} = V_{\text{in}} + 2e^{-2r}\mathbb{I}$ in the ensemble. This optical protocol is named separately from canonical CZ cluster propagation.

6.3 Single-mode symplectic decomposition

Let $S = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $AD - BC = 1$. If $C \neq 0$, the chronological list

$$(k_1, k_2, k_3, k_4) = \left(0, \frac{1-D}{C}, C, \frac{1-A}{C}\right) \quad (23)$$

satisfies $S = T(k_4)T(k_3)T(k_2)T(k_1)$. Multiplication gives the desired A, C, D , with B fixed by determinant one. If C is close to zero, prepend $T(0)$ and apply the same decomposition to $ST(0)^{-1}$, whose lower-left entry is nonzero in the exact $C = 0$ case. The implementation verifies reconstruction and rejects numerically unstable requests. It does not minimize the number or magnitude of shear parameters.

The circuit API supports numeric rotations, squeezing and arbitrary real single-mode symplectic matrices. Compilation emits preparation, CZ, measurement and displacement commands; a direct squeezing gate is not inserted in place of its MBQC implementation. A displacement circuit gate uses identity transport followed by its requested output correction. Hand-written patterns and wire shears retain symbolic parameter support even though the general matrix factorization is performed numerically at compile time.

6.4 Two-mode gates

A logical CZ inserts the corresponding resource coupling between current wire frontiers, then transports both outputs through identity gadgets. The resource edge implements the entangling interaction and the measurements move the logical outputs. Finite-resource noise is retained rather than subtracted in execution.

The logical SUM gate follows by Fourier conjugation on its target. In chronological order, target rotation $R(\pi/2)$, CZ of weight g , and target rotation $R(-\pi/2)$ implement $q_t \mapsto q_t + gq_c$, $p_c \mapsto p_c - gp_t$. Let $U(a)$ and $L(b)$ be the upper and lower shear matrices acting on the two position coordinates. A beam splitter’s real mixing is

$$U(-\tan(\phi/2))L(\sin \phi)U(-\tan(\phi/2)). \quad (24)$$

The compiler uses two half-angle versions to avoid the tangent singularity at $\phi = \pi$. Each shear is a SUM gadget whose local Fourier transformations are themselves measurement-based. This decomposition is correct but expensive; specialized lower-noise two-mode gadgets are a future optimization.

7 Causality, CV-flow and pattern manipulation

CV-flow has a distinct real-linear formulation [5]. To keep this meaning separate from ordinary causal scheduling, `cvflow.py` exposes a certificate for a supplied total measurement order. For each measured node j , let $P(j)$ denote the prefix including j . With input set I , construct

$$A_j = A[P(j), V \setminus (P(j) \cup I)]. \quad (25)$$

The certificate solves $A_j c_j = e_j$, where e_j is one on the current row and zero on earlier rows. It records coefficients and residuals rather than solving over $\text{GF}(2)$.

For each measured value m_j , the generated protocol applies $X(-c_j m_j)$ and the surviving components of $Z(-A c_j m_j)$ on future modes. Inputs are excluded from correction columns because they do not supply the required resource stabilizers. The certificate concerns the specified order; failure does not prove that another order is impossible. Its ideal-limit interpretation does not remove finite-squeezing noise. The implementation is not the full optimal-order finding algorithm from the literature.

7.1 Displacement frames

Pending phase-space translations can be tracked without immediately executing native displacements. If the physical state is represented as a displacement of a stored state by f , a Gaussian gate updates $f \mapsto S f$. Attenuation of transmissivity η sends $f \mapsto \sqrt{\eta} f$. Homodyne shifts its reported value by $h f$ and general-dyne by f ; survivor frames remain pending. Frames are materialized at outputs and before non-Gaussian operations. Seeded physical and frame-based trajectories are compared in regression tests.

7.2 Conservative standardization

`Pattern.standardize` moves preparation and entanglement commands only past disjoint operations and only when classical dependencies permit the swap. It combines adjacent displacements and removes numerical zero shifts. Since displacement ordering can introduce only a global phase, the represented density state is preserved. The implementation does not promise a complete preparation–entanglement–measurement–correction normal form for arbitrary adaptive patterns. In particular, it does not import qubit signal parity, Pauli measurement deletion, or local-Clifford graph rules as CV identities.

8 Independent Gaussian analysis and observables

For a fixed-angle homodyne pattern with affine feed-forward, a separate analyzer propagates lin-

ear Wigner variables. It tracks coefficients of independent input, resource and detector variables, eliminates measured modes from the active map, and adds affine correction coefficients to survivors. This produces

$$\boldsymbol{\mu}_{\text{out}} = S\boldsymbol{\mu}_{\text{in}} + \mathbf{d}, \quad V_{\text{out}} = SV_{\text{in}}S^T + N. \quad (26)$$

The analyzer does not run the conditional trajectory backend. Nonlinear outcome dependencies, outcome-dependent gates, Fock operations and general-dyne measurements are currently rejected by this analysis path, although several are supported by trajectory simulation.

Gaussian output helpers provide quadrature means and variances, reductions, photon number, parity and overlap. In these conventions,

$$\langle n_j \rangle = \frac{\text{tr } V_j + \boldsymbol{\mu}_j^T \boldsymbol{\mu}_j - 2}{4}. \quad (27)$$

For n modes, Gaussian overlap is

$$\text{tr}(\rho_1 \rho_2) = \frac{2^n e^{-\frac{1}{2}\Delta^T(V_1+V_2)^{-1}\Delta}}{\sqrt{\det(V_1+V_2)}}, \quad \Delta = \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2. \quad (28)$$

This equals fidelity when at least one state is pure, and is not advertised as mixed-state fidelity otherwise. Gaussian parity is $e^{-\boldsymbol{\mu}^T V^{-1} \boldsymbol{\mu}/2} / \sqrt{\det V}$.

Repeated shots use independent child random seeds. The ensemble covariance is

$$V_{\text{ens}} = \mathbb{E}[V_{\text{cond}}] + \text{Cov}(\boldsymbol{\mu}_{\text{cond}}). \quad (29)$$

The returned Gaussian container is a moment-matched representation. Nonlinear adaptation can produce a non-Gaussian mixture, for which a Gaussian formula applied to this container need not equal the mixture’s actual observable. Per-trajectory observable averages avoid that misinterpretation.

9 Experimental non-Gaussian measurement computations

Version 0.2 extends the same resource–pattern–execution architecture to pure non-Gaussian trajectories. The extension is not a universal non-Gaussian compiler. It provides finite resources, conditional measurements, classical control and resource-assisted transformations whose mathematical maps can be tested. The preimplementation audit is recorded

in `docs/non_gaussian_design.md`; the complete tutorial and capability contract are in `docs/non_gaussian.md`.

9.1 State space, preparation and capabilities

Piquasso’s pure Fock simulator stores the globally truncated space

$$\mathcal{H}_{d,c} = \text{span}\{|n_1, \dots, n_d\rangle : \sum_j n_j < c\}, \quad (30)$$

$$D(d, c) = \binom{d+c-1}{d}. \quad (31)$$

Thus c is not a per-mode cutoff. For example, two modes at $c = 64$ have $D = 2080$ amplitudes. A complex double vector alone requires $16D$ bytes; density matrices and native gate workspaces can be much larger. Execution requires an explicit integer cutoff. A configurable dimension guard defaults to $D \leq 10^6$, with a warning from $D = 10^5$. This guard bounds the vector space requested, not peak process memory.

`FockInput` retains the single-mode normalized-vector interface. `FockSuperposition` adds sparse occupation tuples with explicit amplitudes. Its mode order follows the labels supplied to `PrepareResource`, or the ordered input labels when supplied as a correlated initial state. Shapes, normalization, finiteness, nonnegative integer occupations, uniqueness and cutoff support are checked. Public native pure Piquasso amplitudes can be copied into this description; arbitrary mixed native input is not accepted. Tensor preparation projects onto the global space and records the retained weight before normalization.

The backend advertises named capabilities through `supports(feature)`. The execution preflight rejects non-Gaussian resources or gates on a Gaussian backend, and rejects unavailable noise/measurement features on the pure-Fock backend. Gaussian and non-Gaussian commands can be interleaved in a pattern executed wholly in Fock space. There is no implicit representation switching. Explicit pure Gaussian inputs are converted to Fock preparations; a mixed Gaussian covariance requires an unavailable mixed-input implementation.

9.2 Cats, cubic resources and unitary conventions

An ideal normalized cat has the form

$$|C_\epsilon(\alpha)\rangle = \frac{|\alpha\rangle + \epsilon|-\alpha\rangle}{\sqrt{2(1 + \epsilon e^{-2|\alpha|^2})}}, \quad \epsilon \in \{-1, 1\}. \quad (32)$$

Its number coefficients are proportional to $\alpha^n[1 + \epsilon(-1)^n]/\sqrt{n!}$, hence parity is exactly ϵ . The mean photon number is

$$\langle n \rangle = |\alpha|^2 \frac{1 - \epsilon e^{-2|\alpha|^2}}{1 + \epsilon e^{-2|\alpha|^2}}. \quad (33)$$

Logarithmic factorials and parity-aware rescaling avoid factorial overflow. **CatResource** constructs the projection at the execution cutoff and monitors its retained infinite-state weight. The older **FockInput.cat** instead defines an explicitly normalized finite vector; a cutoff-dependent factory must rebuild that vector during convergence scans. The zero-norm odd cat at $\alpha = 0$ is rejected.

The nonlinear gate conventions preserve the original adapter and Piquasso:

$$C(\gamma) = \exp(i\gamma q^3/6), \quad C^\dagger p C = p + \gamma q^2, \quad (34)$$

$$K(\kappa) = \exp(i\kappa n^2), \quad K|n\rangle = e^{i\kappa n^2}|n\rangle, \quad (35)$$

$$Q(s) = \exp(isq^2/4), \quad Q^\dagger p Q = p + sq. \quad (36)$$

All angles and coefficients use $\hbar = 2$. A desired phase $\exp(i\lambda q^3)$ therefore requires $\gamma = 6\lambda$. **CubicPhaseResource** represents $C(\gamma)S(-r)|0\rangle$, with finite $r \geq 0$. Its untruncated wavefunction is

$$\phi_{\gamma,r}(q) = \frac{e^{-q^2/(4e^{2r})}}{(2\pi e^{2r})^{1/4}} e^{i\gamma q^3/6}. \quad (37)$$

Preparing such a resource through a simulated cubic gate is a resource model, not a proposal for experimentally synthesizing that gate for free. Ancillary-state approaches to weak cubic nonlinearities and adaptive cubic measurements motivate this separation [7, 8]; the implemented map below is independently derived rather than assumed identical to a specific experimental circuit.

9.3 Ideal ladder operations versus heralding

The mathematical updates **PhotonSubtract** and **PhotonAdd** are

$$|\psi\rangle \mapsto \frac{a|\psi\rangle}{\sqrt{\langle n \rangle}}, \quad |\psi\rangle \mapsto \frac{a^\dagger|\psi\rangle}{\sqrt{\langle n \rangle + 1}}. \quad (38)$$

They apply the occupation factors $\sqrt{n}$ and $\sqrt{n+1}$. In the inspected Piquasso 8.0.1 source, **Create/Annihilate** use unit occupation shifts; the adapter therefore does not equate those instructions with these mathematical ladders. The unnormalized squared ladder norm is recorded separately from truncation weight. It is not a physical success probability: $a^\dagger$ is not by itself a trace-nonincreasing Kraus operator. A numerically zero norm ($\leq 10^{-28}$) and addition leaving the supported space are errors.

The **photon_subtraction** pattern instead couples the input to vacuum with a beam splitter and counts the tap. Under the beam-splitter convention used here, the count- k map on an input number state is

$$M_k|n\rangle = \sqrt{\binom{n}{k}} (\sin \theta)^k (\cos \theta)^{n-k} |n-k\rangle. \quad (39)$$

Consequently

$$P(k|n) = \binom{n}{k} \sin^{2k} \theta \cos^{2(n-k)} \theta. \quad (40)$$

Selecting $k = 1$ is physical heralded subtraction. Finite tap angle retains an attenuation factor; the resulting conditional map is not identical to normalized a . No outcome is silently postselected, and the simulator does not interpret a selected trajectory as the number of experimental trials until success.

9.4 Conditional counting and homodyne

For a joint pure state $\sum_{n,s} A_{n,s}|n\rangle|s\rangle$, number detection gives

$$P(n) = \sum_s |A_{n,s}|^2, \quad B_s = A_{n,s}/\sqrt{P(n)}. \quad (41)$$

Sequential destructive detection on multiple modes realizes the joint count distribution through conditional probabilities. Count records are integers; expressions such as $(-1)^n$ and polynomial corrections use the existing safe expression tree and dependency DAG.

Piquasso's inspected pure-Fock homodyne sampler does not return a conditional state. PhotoGraphiQ adds an explicit wavefunction adapter. Define

$$h_n(x) = \frac{\text{He}_n(x)e^{-x^2/4}}{(2\pi)^{1/4}\sqrt{n!}}, \quad (42)$$

where He_n is the probabilists' Hermite polynomial. For $q_\vartheta = q \cos \vartheta + p \sin \vartheta$,

$$\tilde{B}_s(x) = \sum_n A_{n,s} e^{-in\vartheta} h_n(x), \quad (43)$$

$$p_\vartheta(x) = \sum_s |\tilde{B}_s(x)|^2, \quad (44)$$

$$B_s(x) = \tilde{B}_s(x) / \sqrt{p_\vartheta(x)}. \quad (45)$$

The adapter evaluates h_n by its normalized three-term recurrence. The projection is exact within the stored finite expansion. Sampling numerically integrates this nonnegative marginal and inverts its CDF. Absolute and relative quadrature tolerances are 10^{-10} , root tolerance is 10^{-10} , and the interval $[-2\sqrt{c} - 10, 2\sqrt{c} + 10]$ must capture unit mass within 10^{-7} . Failed integration and zero-density postselection are explicit errors.

Fixed measurement outcomes may be supplied for branchwise experiments. `measurement_statistics` distinguishes conditional probability from probability density and records whether selection was explicit. The sum of their logarithms is a joint log likelihood. With homodyne this is a density, not a nonzero probability of an exact real outcome. Finite detector bins require integration. Noisy Fock homodyne, Fock heterodyne and general mixed measurement updates remain unavailable. Gaussian likelihood diagnostics have not been added to the pre-existing moment backend.

9.5 Finite-resource injection and nonlinear feedforward

Let u be the input and a the resource mode. The chronological sequence $R_a(\pi/2)$, $\text{CZ}_{ua}(-1)$, $R_a(-\pi/2)$ realizes inverse SUM: $q_a \mapsto q_a - q_u$ and $p_u \mapsto p_u + p_a$. Measuring the resource position at m leaves the unnormalized input wavefunction

$$\psi(q)\phi(q+m). \quad (46)$$

`resource_injection` encodes this sequence. Cat injection supplies $\phi = C_\epsilon(\alpha)$ and produces a conditional cat-wavefunction filter; it does not define a deterministic unitary cat gate.

For the cubic resource, expand $(q+m)^3 = q^3 + 3mq^2 + 3m^2q + m^3$. The corrections

$$Q(-2\gamma m), \quad Z(-\gamma m^2) \quad (47)$$

remove the outcome-dependent quadratic and linear phases. Up to global phase, the output of

`cubic_injection` is

$$\psi_{\text{out}}(q) \propto \psi(q) e^{i\gamma q^3/6} \exp\left[-\frac{(q+m)^2}{4e^{2r}}\right]. \quad (48)$$

The finite envelope remains physical filtering; it cannot be discarded when reporting gate fidelity. The state remains on the input node, so this is resource-assisted gate injection rather than relocation of the input state. Increasing squeezing broadens the envelope while increasing photon-number cost. The nonlinear correction m^2 is an ordinary classical dependency in the IR.

```
pattern = pg.non_gaussian.cubic_injection(
    gamma=0.3, squeezing=0.2)
study = pg.cutoff_convergence(
    pattern, [36, 48, 64],
    measurement_outcomes={"m": 0.4})
state = study.results[-1].state
print(study.rows[-1]["fidelity_to_previous"])
```

9.6 Output analysis and truncation evidence

Fock snapshots expose pure vectors where available, density matrices, occupation probabilities, reductions, photon number, parity and quadrature moments. For one mode, with $\mu = \langle a \rangle$ and $\nu = \langle a^2 \rangle$,

$$\langle q_\vartheta \rangle = 2\Re(e^{-i\vartheta}\mu), \quad (49)$$

$$\langle q_\vartheta^2 \rangle = 2\langle n \rangle + 1 + 2\Re(e^{-2i\vartheta}\nu). \quad (50)$$

The $+1$ term avoids the spurious top-level boundary error obtained by squaring a quadrature matrix truncated too early. State comparison aligns occupation tuples, not anonymous vector indices. The implemented fidelity and trace distance are

$$F(\rho, \sigma) = \left[\text{tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right]^2, \quad (51)$$

$$T(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1. \quad (52)$$

When one state is pure, fidelity reduces to $\text{tr}(\rho\sigma)$. Ordered output labels must agree; differing cut-offs are padded by occupations.

The single-mode Wigner marginal uses analytic displacement matrix elements:

$$W(q, p) = \frac{1}{2\pi} \text{tr}[\rho D(q + ip)(-1)^n]. \quad (53)$$

Here $D(\beta) = \exp(\beta a^\dagger - \beta^* a)$ and the phase-space amplitude is $(q + ip)/2$. The density integrates to one over $dq dp$. The finite-grid negative volume is $\int \max(-W, 0) dq dp$ over the stated

window; captured mass is reported separately. Window expansion and grid refinement are independent of Hilbert-space cutoff convergence. Tests include the vacuum and $W_{|1\rangle} = (q^2 + p^2 - 1)e^{-(q^2+p^2)/2}/(2\pi)$, whose full negative volume is $2e^{-1/2} - 1$.

Every native gate and tensor preparation records retained norm and population in the last two total-photon shells. A default norm deviation above 10^{-3} raises an error. Smaller deviations above 10^{-8} warn before normalization; boundary population above 0.02 warns independently. These thresholds are configurable diagnostics, not rigorous error bounds. In particular, the native cubic instruction exponentiates the cube of a truncated position matrix and can conserve norm exactly while representing a poorly converged state.

`cutoff_convergence` compares adjacent increasing cutoffs and records norms, boundary populations, moments, fidelity, trace distance, probability L^1 change, dimensions and local times. Differing outcomes mark trajectories incomparable; using the same seed does not fix a branch. A callable pattern factory rebuilds cutoff-dependent resources. No convergence flag is inferred solely from a norm or a single pair of cutoffs.

9.7 Non-Gaussian validation and reproducible results

Independent dense reference operators occur only in `tests/fock`. They construct $a, a^\dagger, n, q, p$, displacement and squeezing exponentials, and the cubic matrix exponential. Tests use nonzero cubic strengths 0.2 and 0.65, nontrivial Kerr phases $\kappa = 0.71$, analytical ladder factors, heralding probabilities, entangled conditional homodyne states and cat parity. Direct native programs separately validate gates, photon-count frequencies and the synthesized injection interaction. A direct SUM Bogoliubov decomposition and Fourier-CZ synthesis differ slightly at finite cutoff: their observed infidelity decreases from about 4.0×10^{-7} at $c = 24$ to 1.8×10^{-8} at $c = 48$ for the stated test resource. Identical native instruction sequences agree to numerical tolerance; different finite-space projections are not required to coincide exactly.

The cat and cubic injection tests integrate Eq. (46) or Eq. (48) independently into Fock co-

Experiment	c	$1 - F$	min. norm
cubic gate	24	4.76e-03	1.000000
cubic gate	48	3.82e-04	1.000000
cubic gate	80	1.48e-05	1.000000
cubic injection	48	1.45e-06	0.999984
cubic injection	64	2.92e-07	0.999997
cat	16	7.53e-06	1.000000
cat	24	4.37e-09	1.000000

Table 2: Reproduced adjacent-cutoff comparisons. The cubic gate acts on vacuum with $\gamma = 0.65$; injection uses $\gamma = 0.3, r = 0.2, m = 0.4$ and a vacuum input; the even cat has $\alpha = 1.5$. Minimum norm is measured before normalization over each trajectory. Fidelity compares to the preceding cutoff, not an ideal infinite-energy gate.

efficients. For a vacuum input, cubic-resource homodyne density is

$$p(m) = \frac{\exp[-m^2/(2(1 + e^{2r}))]}{\sqrt{2\pi(1 + e^{2r})}}, \quad (54)$$

independent of γ before conditioning corrections. The test at $\gamma = 0.3, r = 0.2, m = 0.4, c = 36$ checks this density and conditional wavefunction. Graphix supplies only a corresponding preparation-edge-measurement-correction skeleton and classical domains; its qubit amplitudes never validate these maps.

For the strong cubic gate on vacuum, the untruncated theory gives $\langle p \rangle = \gamma$ and $\langle n \rangle = 3\gamma^2/4 = 0.316875$. At $c = 80$ the recorded mean photon number is approximately 0.316868; norm remains one at every tested cutoff. This is numerical evidence on the specified state, not a uniform bound for stronger gates or arbitrary inputs. For finite injection the adjacent $48 \rightarrow 64$ fidelity is approximately 0.999999708, while the residual envelope in Eq. (48) remains.

The new production modules are `resources` (cutoff-aware descriptions), `capabilities` (pre-flight), `fock_measurements` (conditional homodyne), `non_gaussian` (resource patterns), `fock_analysis` (metrics and Wigner grids), and `convergence` (branch-aware scans). The existing commands, states, serialization, simulator and Fock backend are extended without replacing Gaussian compilation or channel analysis.

9.8 Noise and present boundaries

Mixed Fock inputs, Fock loss channels, noisy Fock homodyne, general non-Gaussian POVMs,

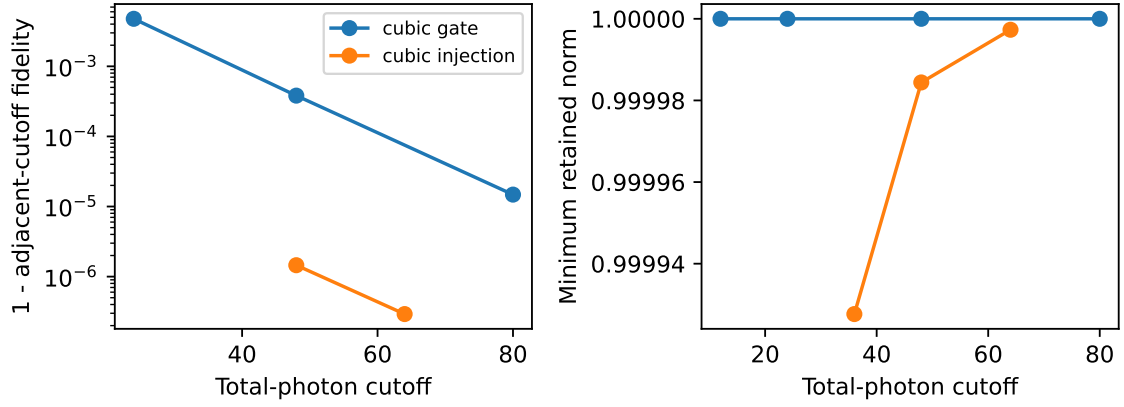


Figure 2: Cutoff dependence distinguishes a norm-preserving truncated cubic unitary from converged output physics. The injection trajectory is fixed at $m = 0.4$; its residual norm loss and state change both decrease. Numerical values and local timings are stored in `results/non_gaussian.csv`.

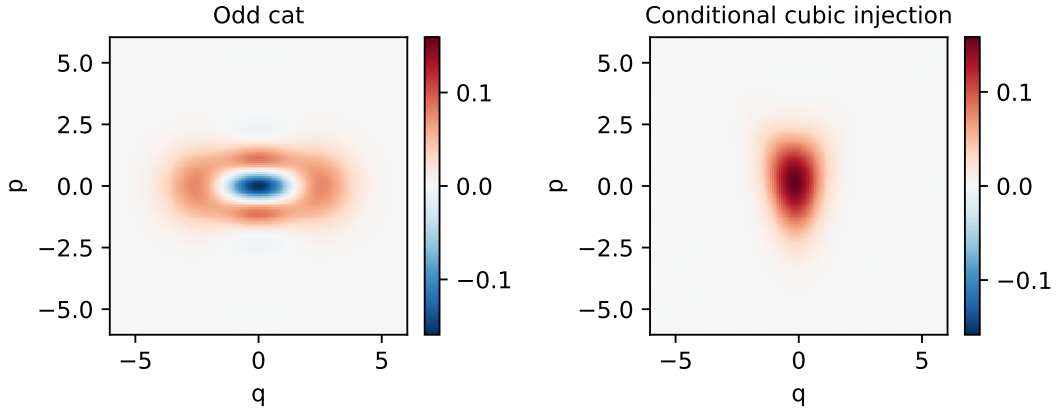


Figure 3: Single-mode Wigner densities for an odd cat ($\alpha = 1.2, c = 24$) and the fixed cubic-injection output ($c = 64$). The 161×161 window is $[-6, 6]^2$. Captured masses and finite-window negative volumes are saved in `results/non_gaussian_wigner.json`. These plots are numerical diagnostics; their finite-grid negativity is not a hardware certification.

dynamic Gaussian-to-Fock switching, GKP decoding and a universal non-Gaussian compilation path are outside this release. The implemented operations should be described as experimental non-Gaussian MBQC support. Gaussian noise support remains thermal attenuation, $V \mapsto XVX^T + Y$, with local $X = \sqrt{\eta}\mathbb{I}$ and $Y = (1 - \eta)(2n_{\text{th}} + 1)\mathbb{I}$. Detector inefficiency is a Gaussian measurement parameter and finite squeezing is a resource property. Phase diffusion, mode mismatch and detailed detector hardware are not silently approximated by these channels.

10 Validation and reproducible results

Validation uses three distinct sources of evidence. Analytical tests establish vacuum conventions, symplectic transformations, weighted-resource nullifiers, Schur conditioning, output-channel matrices, detector statistics, and CV-flow equations. Raw Piquasso tests compare explicit photonic programs with the adapter, including native conditional moments and a separately identified sampler-convention characterization. GraphiX tests compare connectivity, input/output designation, command/measurement order, correction domains and safe preparation standardization. No test requires CV and qubit state equality.

The two Gaussian backends share the mea-

r	Exact V_q	Sample V_q	Sample V_p	Exact F
0.3	2.0976	2.0255	2.1505	0.6457
0.6	1.6024	1.5775	1.5435	0.7685
0.9	1.3306	1.3279	1.3229	0.8581
1.2	1.1814	1.1789	1.1647	0.9168
1.5	1.0996	1.0976	1.0925	0.9526

Table 3: Four-step identity-wire study, 600 shots per row. Variances are in units where vacuum variance is one. Samples include both conditional covariance and covariance of conditional means.

surement adapter; their agreement alone does not independently validate measurements. Independent analytic conditioning, statistical tests and the Wigner channel analyzer address this limitation. Deterministic comparisons use tight numerical tolerances on modest states, whereas finite-shot comparisons use looser fixed-seed statistical tolerances. The validation artifact `results/tests.xml` records the executed suite.

10.1 Finite-squeezing identity wire

We simulate a four-step identity wire with vacuum input at five resource squeezing values. Each point uses 600 independent Gaussian trajectories and seed base 20260910. Equation (22) predicts

$$V_{\text{out}} = (1 + 2e^{-2r})\mathbb{I}. \quad (55)$$

For a coherent input, displacement covariance implies the same added noise, and the overlap with the ideal pure coherent target is

$$F(r) = \frac{1}{1 + e^{-2r}}. \quad (56)$$

Table 3 contains the actual generated covariance estimates and the analytic coherent-state fidelity. The fidelity column is a prediction, not a finite-shot fidelity estimate. The maximum absolute covariance-entry error in this saved 3,000-trajectory study is below 0.073.

10.2 Reproduction procedure

The repository contains package metadata, a frozen local dependency list, CI configuration, unit and integration tests, tutorials, example scripts, a figure generator, source manuscript and compiled figures. The reference environment uses Python 3.12, Piquasso 8.0.1 and GraphiX 0.4. The generated environment file records precise

NumPy, SciPy, NetworkX and plotting versions. Local CPU timings in the CSV are provenance only; they are not cross-platform benchmarks.

```
import photographiq as pg

p = pg.Circuit(1).rotate(0, 0.4).squeeze(0, 0.2)
pattern = p.compile(squeezing=1.2)
result = pg.simulate(
    pattern,
    inputs={0: pg.GaussianInput.coherent(0.3)},
    seed=42, frame=True,
)
channel = pg.gaussian_channel(pattern)
print(result.outcomes)
print(channel.matrix, channel.noise)
```

Install the development extras and run `python -m pytest`. Running `python experiments/reproduce.py` regenerates the CSV, environment JSON, LaTeX table and figures. Running `python experiments/non_gaussian.py` regenerates the non-Gaussian convergence CSV, figures and local environment record. The manuscript uses the official Quantum document class, included with its original license. Running pdfLaTeX twice in the paper directory builds the manuscript. No remote quantum service or hardware account is required.

11 Limitations and future development

The implementation favors explicit physics and auditability. Dense covariance storage requires quadratic memory, and full physicality checks use dense eigensolvers. The native adapter also copies moments between calls. Streaming teleportation keeps the active state small, but arbitrary large graph preparation does not yet exploit sparse covariance or specialized structure. No superiority over optimized Piquasso execution is claimed.

The Gaussian compiler is intentionally conservative and can use many ancillas, especially for beam splitters. Its finite-resource noise may be substantially larger than that of specialized optical gadgets. Optimizing resource count and noise jointly is a future task. The supplied-order CV-flow implementation does not solve the full order-search problem. The standardizer is not a complete CV rewrite calculus.

The stochastic execution engine converts scalar parameters to Python floats. Symbolic construc-

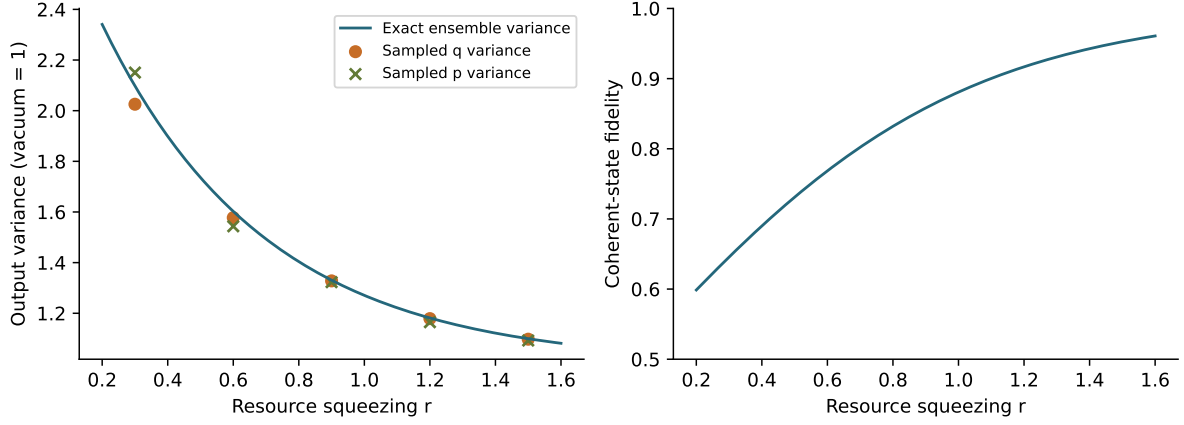


Figure 4: Reproducible finite-resource identity transport. Left: the exact unconditional output variance and two sample estimates. Right: analytic coherent-state fidelity of the noisy identity channel. Curves are calculated from the derived channel; markers are generated by independent trajectories. These results are simulation data, not photonic hardware measurements.

tion and upstream differentiable connectors do not make it end-to-end differentiable. Future PhotoGraphiQML work would require a dedicated gradient-aware execution contract, estimator choices and batching support. Gaussian fitting of adaptive mixtures must not obscure non-Gaussian ensemble features.

Non-Gaussian resource experiments remain bounded by the explicit Fock cutoff. Noisy Fock homodyne, mixed-state input resources, GKP error correction, fault-tolerant compilation and realistic temporal hardware scheduling are not implemented. These omissions are exposed as documented capability boundaries rather than apparently functional placeholder APIs.

12 Conclusion

PhotoGraphiQ provides a tested computational layer for expressing, compiling, executing and analyzing finite-squeezing CV measurement-based computations. Its software abstractions distinguish resource topology, measurement calculus, classical causality and physical simulation. Independent conventions and channel analysis make finite-resource effects visible, while raw Piquasso and structural GraphiX tests answer different validation questions. This separation provides a practical basis for extending Gaussian MBQC studies and carefully bounded non-Gaussian resource experiments.

Code and data availability

Original software is released under the MIT license at <https://github.com/chinmoybiswasdeep/PhotoGraphiQ>. The local repository contains all source and research artifacts described here; this manuscript does not assert that the generated changes have been pushed or released remotely. Reproduction data are in `paper/results`; research PDFs supplied for study retain their original copyrights. The Quantum class retains its LaTeX Project Public License. No software DOI is assigned in this draft.

Author contributions and AI-use disclosure

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A Backend protocol and extension boundaries

`BaseBackend` specifies reset, preparation, entanglement, displacement, rotation, squeezing, beam splitting, destructive measurement and state retrieval. Optional loss and cubic-phase operations have explicit unsupported defaults. A backend instance passed to `simulate` is reset for each trajectory. State interfaces use physical labels and

the conventions in Sec. 3, independently of native mode indices.

The Piquasso adapter constructs native Gaussian states from public moment properties, executes native instruction programs and converts moments back. External correlated native states can be imported with an explicit source $\hbar$, rescaling means by $\sqrt{2/\hbar_{\text{source}}}$ and native anticommutator covariance by $1/\hbar_{\text{source}}$. This avoids treating a vacuum in one convention as a squeezed or unphysical state in another.

Future backends must preserve destructive measurement semantics, maintain labels after mode removal, implement deterministic seed handling, and expose unsupported operations honestly. New covariance or Fock representations should not leak into graph and expression modules. New optimization passes require equivalence tests rather than assumptions based on qubit command names.

B Interchange, visualization and repository checks

Pattern JSON is tagged with a schema name and version. Its decoder only constructs an allowlist of commands, measurement descriptions, input resources and expression nodes. Labels and tuples preserve their types. Nonfinite JSON constants and unknown expression operations are rejected; callables are runtime-only. The format stores computations rather than stochastic trajectories.

Resource drawings display input/output colors, labels, squeezing and edge weights. Pattern drawings add ordered measurement annotations. Dependency drawings show the command DAG, including quantum ordering as well as classical dependencies. These figures are standard matplotlib artifacts suitable for export; they are not diagrams of physical interferometer layouts.

CI configuration covers Python 3.11, 3.12 and 3.13 and includes pytest, Ruff and wheel/source-distribution builds. Static checking uses Python 3.12 because the installed recent NumPy type stubs use its syntax. The local run is one tested environment, not evidence that every configured CI platform has already passed. Source formatting, annotations, a pre-commit configuration and contribution instructions support later extension. The supplied upstream PDFs remain

study materials, and local source checkouts are excluded from distributions.

C Test interpretation and numerical caution

The raw native Gaussian sampling characterization is intentionally distinct from a physics-correctness test. It pins an observed dependency behavior so that an upstream change triggers review of the adapter and documentation. It should not be read as endorsing an incorrect covariance convention. Native gate, graph and conditional-state tests compare the properties they actually exercise.

Physicality checks use finite tolerances and cannot substitute for conditioning analysis at extreme squeezing. The compiler checks symplectic reconstruction, but large coefficients can still amplify finite-resource noise. CV-flow least-squares residuals are numerical certificates for modest, well-scaled examples; users should inspect coefficients and sensitivity for ill-conditioned weighted graphs. A rejected order can be retried with another order, but the software does not silently claim a global impossibility result.

Finally, normalization of a truncated Fock state preserves its use as a finite model but does not establish convergence to an infinite-dimensional state. Studies involving cubic phase or high squeezing should compare observables and probabilities over increasing cutoffs and report retained norms. The Gaussian path remains independent of this truncation and should be preferred whenever the computation remains Gaussian.